

To the Editors
Intelligent Systems with Applications

Re: Original research submission — “Evaluating LLM-Based Credit Rating Systems: A Pre-Registered Specification-Curve Analysis.”

Dear Editors,

We are pleased to submit the above manuscript for consideration in *Intelligent Systems with Applications*. The paper contributes an intelligent-systems reliability framework for evaluating the model risk of using large language models as credit-rating systems. Using a pre-registered specification-curve design, we measure whether an LLM-generated credit rating is stable when the underlying financial information is held fixed but reasonable elicitation choices vary. Across a fixed 90-item battery crossed with a balanced grid of model version, temperature, prompt paraphrase, output format, few-shot exemplars, and presentation across two providers (8,640 elicitations, four model snapshots), no controllable elicitation axis explains more than about 5% of rating variance, pure run-to-run seed variation is only about 8%, the instability survives verified deterministic execution (permutation $p = 0.001$), and 25.2% of matched elicitation pairs disagree at the investment-grade/high-yield boundary. We translate this into decision-relevant units—53.5% implied portfolio turnover and a \$2.0–4.1bn Pillar-1 capital range on the same \$45bn book—and show the pattern persists on 31 decontaminated real issuers under a pre-registered fingerprinting gate.

The work fits the journal’s scope—intelligent systems applied to business and finance—and speaks directly to the deployment of LLMs in financial decision support. Beyond the diagnosis, we propose a deployable control: a *specification-instability score*, computed from the model’s own outputs, that can be reported alongside any LLM credit rating so that specification sensitivity is characterized rather than hidden in a single point estimate. Methodologically, the study is pre-registered, uses issuer-clustered resampling and permutation inference for its main uncertainty assessments, and provides a deterministic offline reproducibility capsule (Zenodo, <https://doi.org/10.5281/zenodo.21953935>; MIT license) that regenerates every reported number, table, and figure. The study uses only synthetic firm profiles and publicly available SEC filings, involves no human or animal subjects, and discloses its use of generative-AI assistance.

The manuscript is original, is not under consideration elsewhere, and has been approved by both authors, who declare no competing interests and received no external funding. Thank you for your consideration.

Sincerely,

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